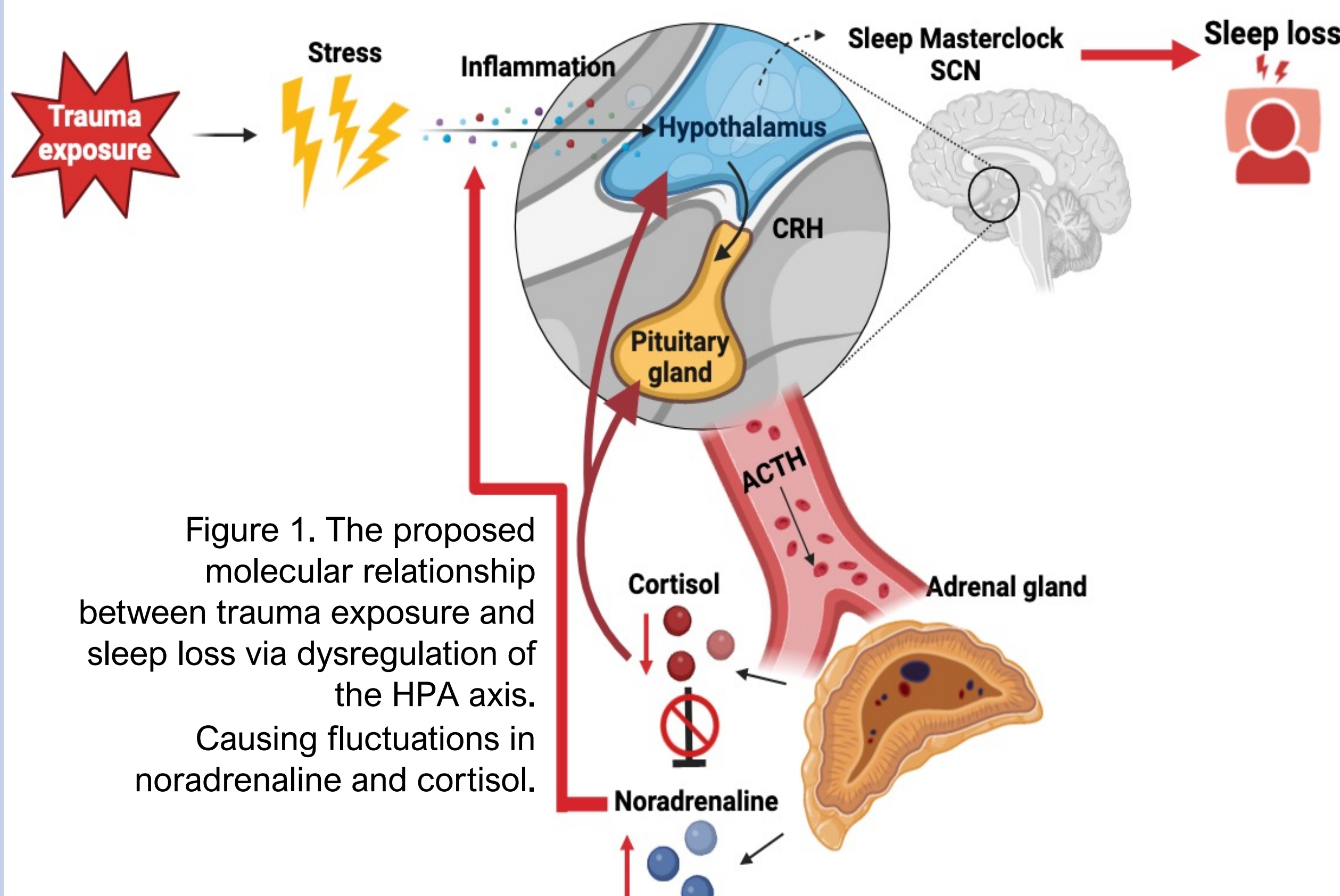


Introduction

Traumatic events trigger stress responses, causing different physical and psychological reactions as the body tries to restore homeostasis.

Long-term, this can lead to impaired health and quality of life. Stress and sleep loss due to trauma exposure has been attributed to noradrenergic and cortisol fluctuations via the hypothalamic-pituitary-adrenal (HPA) axis [1] (Figure 1)



Childhood trauma exposure increases the risk of sleep disturbances across the lifespan [2].

Adult trauma exposure, within the context of combat, is linked to causing insomnia as part of post-traumatic stress disorder diagnoses, with less data on wider populations [3].

The UK BioBank [4] contains biomedical data on 502,357 adults (ages 37 – 73), including trauma exposure and self-reported sleeplessness/insomnia data on 157,298 participants.

Providing the largest databank of adults with and without trauma exposure within the UK.

Research gap: Exploring trauma exposure in childhood and adulthood in a large adult cohort from the general population. Providing clearer understanding of the long-term effects of trauma exposure on sleep health.

Hypothesis

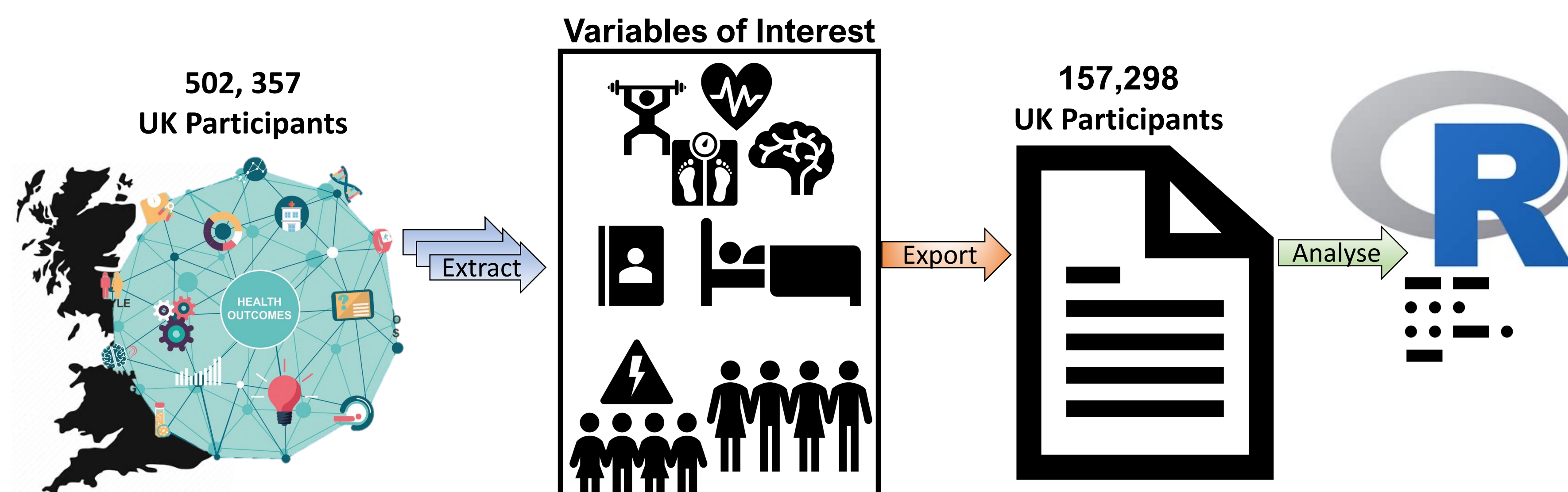
Exposure to trauma in childhood, adulthood (and during both periods) will be associated with increased risk of insomnia in adulthood.

Aims

1. Impact of age of trauma exposure on adulthood insomnia using the scores of the Thoughts and Feelings questionnaire and self-reported sleeplessness/insomnia data in a UK BioBank cohort. Comparator groups will demonstrate risk.
2. Explore biological sex and subscales of abuse and neglect in relation to insomnia.

Research Plan

Trauma variables (predictor), sleeplessness/insomnia (outcome) and sociodemographic data (confounding) are extracted (Figure 2) from the BioBank.



biobank^{uk} Thoughts and feelings

Childhood

Q1 When I was growing up...

a) I felt loved

☐ Never true
☐ Rarely true
☒ Sometimes true
☐ Often
☐ Very often true
☐ Prefer not to answer

b) People in my family hit me so hard that it left me with bruises or marks

☐ Never true
☐ Rarely true
☒ Sometimes true
☐ Often
☐ Very often true
☐ Prefer not to answer

c) I felt that someone in my family hated me

☐ Never true
☐ Rarely true
☒ Sometimes true
☐ Often
☐ Very often true
☐ Prefer not to answer

d) Someone molested me (sexually)

☐ Never true
☐ Rarely true
☒ Sometimes true
☐ Often
☐ Very often true
☐ Prefer not to answer

e) There was someone to take me to the doctor if I needed it

☐ Never true
☐ Rarely true
☒ Sometimes true
☐ Often
☐ Very often true
☐ Prefer not to answer

Figure 3. The 5 variables of the Childhood Trauma Screener (CTS). Likert-style scale: 0-4 "Never true" – "Very often true"

10 trauma variables:

5-item Childhood Trauma Screener

A bespoke 5-item adulthood screener.

Excluding variables with ambiguous timeframe e.g. lifetime.

Do you have trouble falling asleep at night or do you wake up in the middle of the night?

Self-reported sleeplessness/insomnia measured by self-report Likert-scale.
Never/rarely = 1
Sometimes = 2
Usually = 3

R Data analysis: Ordinal logistic regression model: Sleeplessness/insomnia as the outcome measure is scored from 1-3 (ordered, categorical). Identify multiple collinearity between confounding variables (ethnicity, TDI, smoking, BMI), sleeplessness/insomnia and trauma variables, adding to regression.

Preliminary Results

	No Trauma (control) N = 31,186	Childhood Trauma N = 24,579	Adulthood Trauma N = 31,871	Both Child/Adulthood Trauma N = 69,662*
Mean Age (years)	56 ± 8	56 ± 8	56 ± 8	56 ± 8
Sex (%)				
Female:	55.1	48.1	61.0	58.3
Male:	44.9	52.9	39.0	41.7
Ethnicity (%)				
British:	94.16	92.38	91.11	88.14
Irish/White:	4.43	5.33	5.92	7.17
Other:	0.80	1.46	1.90	3.00
Mean trauma score (composite)	0	2.36 ± 1.92	3.01 ± 2.11	6.96 ± 4.3

Table 1. Summary age, sex and ethnicity and mean trauma composite score (out of 20, out of 40 for mixed group) data for the four experimental groups; participants with no trauma (control); only childhood trauma; only adulthood trauma and both child and adulthood trauma. Mean ± SD. Statistics calculated using R.

Conclusions

- Trauma exposure and associated stress are known to impact sleep and circadian rhythm
- The UK Biobank dataset is a rich dataset to explore the impacts of trauma exposure in childhood and adulthood on sleep
- Preliminary analysis reveals mean age of 58 for each cohort and similar ethnic groupings but differ in gender majority. The mixed cohort is 2 – 2.5x larger*
- Future analyses will investigate the relationship between trauma exposure and insomnia

Limitations

- Heterogeneity unaccounted for in severity of the traumatic event on a 5-point Likert scale
- Recall bias in trauma answers
- BioBank population ≠ broader UK population

Future Directions

There are many more trauma and sleep variables in the BioBank that are outside the scope of this current project:

Do you consider yourself to be?

☐ Definitely a 'morning' person

☐ More a 'morning' than 'evening' person

☐ More an 'evening' than a 'morning' person

☐ Definitely an 'evening' person

☐ Do not know

☐ Prefer not to answer

Chronotype is a very useful measure of circadian sleep/wake cycles.

More of the Thoughts and Feelings questionnaire covers Lifetime trauma such as war/combat and life-threatening diagnoses and accidents, of keen interest within sleep loss research.

Acknowledgments

I would like to thank my postdoctoral supervisors Dr Amber Roguski and Raahat Manrai and Professor Daniel Smith for their ongoing guidance and support.

References

- [1] Lo Martire, V., et al. (2020). Stress & sleep: A relationship lasting a lifetime. *Neuroscience and biobehavioral reviews*, 117, 65–77. <https://doi.org/10.1016/j.neubiorev.2019.08.024>
- [2] Desch, J., et al. (2023). The association between adverse childhood experiences and insomnia symptoms from adolescence to adulthood: Evidence from the Add Health study. *Sleep health*, 9(5), 646–653. <https://doi.org/10.1016/j.sleh.2023.06.001>
- [3] Short, N. A., et al. (2022). The association between traumatic life events and insomnia symptoms among men and women: Results from the Baltimore Epidemiologic Catchment Area follow-up study. *Sleep health*, 8(2), 249–254. <https://doi.org/10.1016/j.sleh.2021.11.008>
- [4] This research has been conducted using data from UK Biobank, a major biomedical database. www.ukbiobank.ac.uk